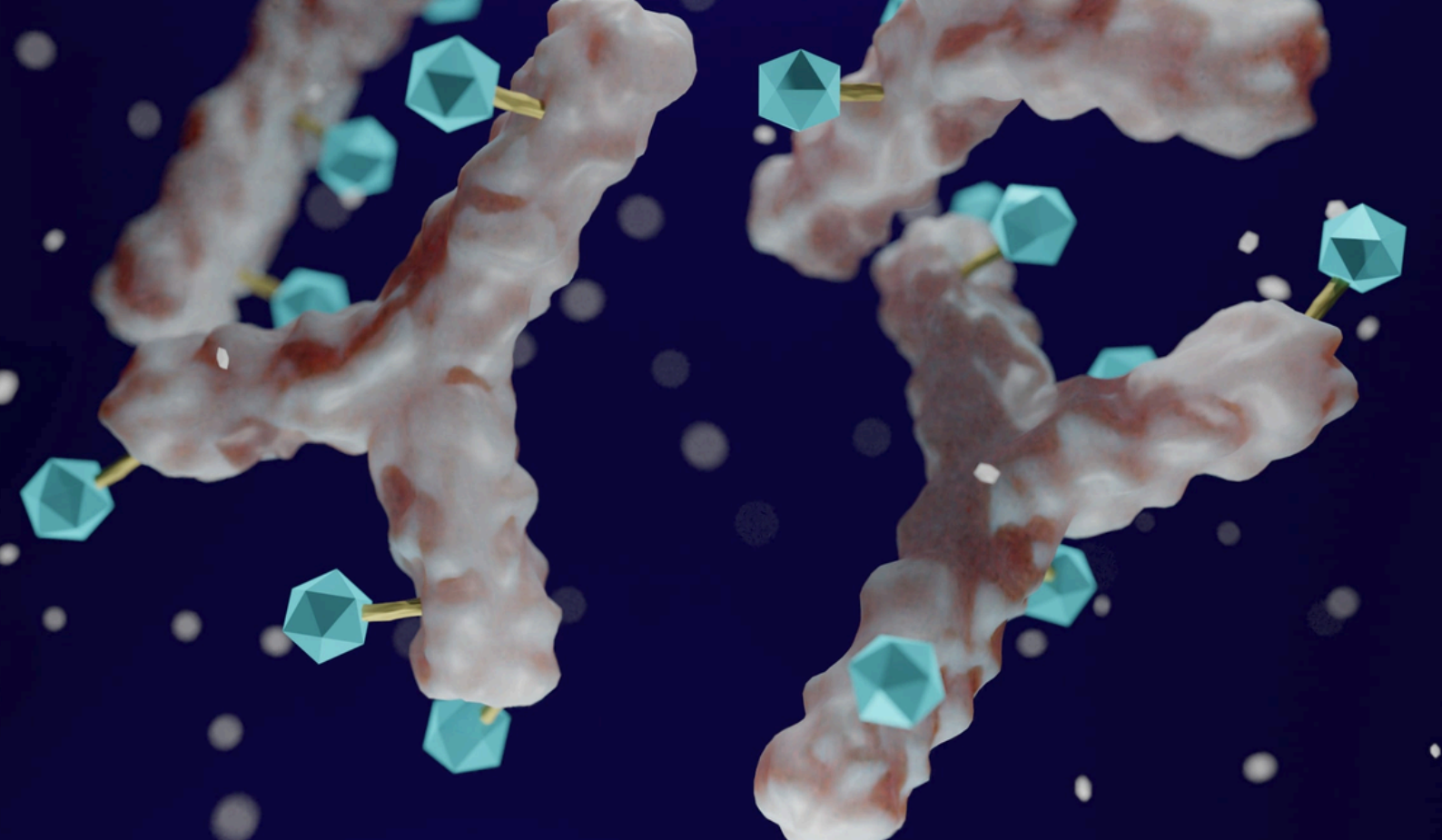




**FEBRUARY 2025 (TRACK 2- TREATMENT AND
DRUG DISCOVERY)**



DeepCure

**AI-Driven Antibody-Drug Conjugate (ADC)
Design for Cancer Therapy**

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Problem Statement

Antibody-drug conjugates (ADCs) have emerged as a paradigm-shifting innovation in oncological therapeutics, offering targeted cytotoxic delivery with reduced systemic toxicity. However, the rational design and optimization of ADCs remain formidable challenges due to the multifactorial interplay of antigen selection, antibody specificity, linker stability, payload potency, and the drug-antibody ratio (DAR). Conventional ADC development is predominantly reliant on high-throughput screening and empirical optimization, both of which are labor-intensive, cost-prohibitive, and constrained by experimental throughput.

Existing computational approaches exhibit intrinsic limitations that hinder their predictive reliability. Classical molecular docking strategies inadequately capture the dynamic nature of ADCs, while many machine learning models fail to effectively integrate heterogeneous biological, chemical, and structural datasets. Moreover, the lack of a unified, multi-modal computational framework that synergistically incorporates antigen-antibody interactions, linker chemistry, payload toxicity, and three-dimensional conformational biology remains a critical gap in the field.

To address these deficiencies, an advanced AI-driven framework is imperative for the accurate prediction of ADC efficacy, enabling rational design strategies, and accelerating the development of next-generation, patient-specific ADC therapies.

Proposed Solution

This project proposes a sophisticated multi-modal deep learning architecture that augments the predictive capabilities of ADCNet by integrating state-of-the-art artificial intelligence methodologies:

Protein Representation Learning

To enhance the precision of antigen-antibody interaction modeling, this framework will employ ProtT5-XL, OmegaFold, and ESM-2, facilitating the extraction of high-dimensional sequence embeddings. Additionally, AlphaFold-derived three-dimensional protein structures will be incorporated to refine binding site predictions and improve the accuracy of docking simulations.

Molecular Representation & Graph Neural Networks (GNNs)

The molecular properties of ADC linkers and payloads will be encoded using advanced graph-based neural architectures such as MPNN and SchNet, allowing for a more nuanced understanding of molecular interactions and stability. Hybrid graph-sequence networks will be deployed to model linker degradation kinetics and payload activation dynamics.



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Proposed Solution

Unified Multi-Input AI Model

A fusion network will be designed to integrate multiple data modalities, encompassing sequence-based (antigen-antibody interactions), graph-based (linker-payload chemistry), and structural (3D docking) inputs. An attention-weighted fusion network will prioritize critical features, enhancing the **accuracy of ADC efficacy and toxicity predictions.**

Web-Based Interactive Platform

The AI model will be embedded within a cloud-based interactive interface, enabling real-time ADC design optimization. Researchers will be able to input molecular structures and receive predictive insights on ADC stability, cytotoxic potential, and therapeutic index. The platform will also suggest structural modifications to enhance ADC performance.

By leveraging this multi-disciplinary, AI-enhanced approach, the proposed framework offers a scalable, cost-efficient, and highly accurate strategy for ADC development, expediting the transition from computational design to preclinical validation.

Applications & Impact

High-Throughput Drug Discovery & Preclinical Prioritization

The implementation of this AI-driven ADC discovery platform will significantly accelerate drug screening by computationally prioritizing high-potential candidates. By predicting antigen compatibility and linker-payload integrity, this system will mitigate the costs associated with experimental failures and streamline the preclinical pipeline.

Precision Oncology & Personalized Treatment

By integrating patient-specific genomic and proteomic data, this framework will facilitate personalized ADC therapies. Predictive modeling of patient-specific antigenic landscapes will enable the rational selection of ADCs tailored to individual tumor profiles, thereby optimizing therapeutic efficacy and minimizing adverse effects.

Mechanistic Insights & Resistance Profiling

The systematic analysis of ADC performance using AI will illuminate novel biological mechanisms underpinning ADC activity and resistance. This knowledge will guide the rational design of next-generation ADCs and assist in identifying resistance-associated biomarkers for combination therapy strategies.



Applications & Impact

Multi-Modal Synergistic Drug Regimens

The AI framework will extend beyond standalone ADC prediction to evaluate the potential efficacy of ADCs in combination with other therapeutic modalities, such as immune checkpoint inhibitors, kinase inhibitors, and chemotherapeutic agents. This will facilitate the identification of synergistic drug regimens, thereby expanding ADC utility across diverse oncological indications.

Challenges & Future Prospects

The effectiveness of this model will be contingent on the availability of high-quality ADC datasets, necessitating ongoing curation and integration of expanding experimental repositories. Model generalizability will require continuous retraining with novel ADC structures to prevent overfitting to existing molecular frameworks. Furthermore, to bridge the gap between in silico predictions and clinical translation, prospective validation using large-scale patient-derived datasets will be essential.

Interdisciplinary Collaboration

The successful execution of this project necessitates a confluence of expertise spanning multiple scientific domains. Oncologists and molecular biologists will provide empirical insights into antigenic heterogeneity and ADC pharmacodynamics. Bioinformaticians will curate and preprocess high-dimensional genomic and proteomic datasets. Computational chemists and chemoinformatics specialists will optimize molecular embeddings for linker and payload characterization, while AI and machine learning engineers will architect the deep learning frameworks. Finally, pharmaceutical scientists and translational researchers will conduct experimental validation in preclinical and clinical settings.

This integrative collaboration will establish a transformative AI-powered paradigm for ADC discovery, fundamentally reshaping the landscape of precision oncology.

